

Self-Supervised Representation Learning for Detecting Artificial Intelligence-Generated vs Real Images with Latent Space Analysis

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing a degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. All main sources of help have been acknowledged.

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Abstract

Artificial Intelligence image generators can now create realistic pictures of people, places, objects, and artwork. These images can support creative work, but they can also be used for misinformation, identity misuse, false evidence, and online fraud. This makes real-versus-generated image detection an important task in digital image forensics.

Many existing detectors work well when the training and test images come from the same generator. The problem becomes harder when the test image comes from an unseen generator, a different image domain, or a compressed online version of the same image. This report therefore focuses on a detector that is less dependent on one known generator and more focused on the learned image representation.

The proposed study uses self-supervised or contrastive representation learning to learn image embeddings from real photographs and generated images. The planned image sources include GAN-based and diffusion-based generators. The encoder is then frozen and the embeddings learned during training are analyzed with PCA, UMAP and t-SNE. Real and generated images are separated by Silhouette Coefficient, AUC, accuracy, F1-score and average precision.

The novelty of work is maintained in a practical manner. The study is not designed to develop a very large new model. Instead, it integrates four parts that can be implemented: self-supervised feature learning, cross-generator testing, latent-space explanation and robustness testing under JPEG compression and resizing. This introduces a direction to the thesis research, going beyond mere binary classification.

Keywords: Artificial Intelligence-Generated Images; Synthetic Image Detection; Self-Supervised Learning; Contrastive Learning; Latent Space Analysis; Cross-Generator Generalization; Digital Image Forensics; PCA; UMAP; t-SNE; Silhouette Coefficient.

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Nomenclature

The following is a list of symbols and abbreviations that will be used in the body of the document

AI Artificial Intelligence

AUC Area Under the Receiver Operating Characteristic Curve

BYOL Bootstrap Your Own Latent

CLIP Contrastive Language-Image Pre-training

CNN Convolutional Neural Network

DIRE Diffusion Reconstruction Error

DRCT Diffusion Reconstruction Contrastive Training

EXIF Exchangeable Image File Format Metadata

GAN Generative Adversarial Network

MLP Multilayer Perceptron

PCA Principal Component Analysis

SelfCon Self-Contrastive Learning

SSL Self-Supervised Learning

t-SNE t-distributed Stochastic Neighbor Embedding

UMAP Uniform Manifold Approximation and Projection

WebP Web Picture Image Format

Chapter 1

Introduction

1.1 Background

Cameras aren't the only ones that make images. Modern generative models can generate realistic images from a text-based description, a latent vector or an editing instruction. The first generation of GAN-based faces and scenes, including ProGAN, StyleGAN and BigGAN, demonstrated that artificial faces and scenes can be extremely realistic [1].

The problem has been made more general by the diffusion and text-to-image systems. Systems like Stable Diffusion, GLIDE, Dall-e style systems, Midjourney and SDXL can generate pictures that could appear natural to average viewers. Studies on text-to-image detection and diffusion detection show that these models leave different traces from earlier GAN systems [2], [3].

This creates a direct challenge for digital image forensics. A generated image can be posted online, compressed, resized, cropped, and shared again before verification is done. Work on datasets such as GenImage, WildFake, ArtiFact, and ImagiNet shows that a practical detector must handle different generators, content types, and online image conditions [4], [5].

Earlier studies also show that generated images may contain hidden forensic signals. Some signals appear in frequency patterns, some in local pixel relationships, and some in reconstruction behavior. Co-occurrence matrices, Synthbuster, NPR, DIRE, and AEROBLADE each show that a detector may need more than normal RGB-level classification [6], [7].

A common weakness is still present in many methods. A fully supervised detector may learn artifacts from the training generator instead of learning broader image-level differences. Universal fake detector and CLIP-based works show that a broader feature space can improve transfer to unseen generators [8], [9].

Self-supervised representation learning gives a useful direction for this problem. Instead of depending only on real/fake labels, the encoder can learn stable image features first. SimCLR, BYOL, SelfCon-style learning, and camera-metadata-based self-supervision all support the idea that a better feature space can make detection more generalizable [10], [11].

1.2 Rational of the Study or Motivation

The main motivation of this study is the generalization gap in Artificial Intelligence-generated image detection. A detector may show high accuracy on a known gener-

ator but fail when the image comes from a newer model. This weakness is serious because new generators are released frequently, and the source generator is usually unknown in real use [12], [13].

A recent paper also indicates that the feature space is important. The methods can be based on RGB or frequency maps, co-occurrence matrices, reconstruction errors, CLIP embeddings, or prompt-aware representations. These decisions reveal various forensic hints [6], [14].

The second motivation is interpretability. A normal classifier provides a probability but may not provide clarity on how the real and generated images are separated. PCA, UMAP, t-SNE, and Silhouette Coefficient can be used to perform latent-space analysis to determine if real and GAN images, or real and diffusion images, are meaningful groups [15], [16].

The third motivation is for practical deployment. Images found online are sometimes resized or compressed, or converted in some way, prior to detection. The robustness of a method under realistic sharing conditions is now an important consideration in synthetic image detection research, as seen in datasets and benchmarks like ImagiNet and WildFake [5], [17].

1.3 Problem Statement

The research problem can be summarized in five points.

- Overfitting to generator-specific artifacts: Many detectors overfit the artifacts of single generator. Their performance can be excellent on that generator, but not on generators they haven't seen (GAN), diffusion, or commercial generators.
- Strong reliance on labeled fake images: Fully supervised methods require labeled samples of known generators. This will cause the detector to become "reactive" as new generators appear.
- One-sided latent-space explanation: Some papers state the accuracy, the AUC or the average precision, but do not mention what the real and generated samples correspond to in the learned feature space.
- Bias in the dataset and content: If the real and synthetic images are not equally distributed in terms of content, image type or source, then the model can learn differences in categories.
- The robustness of these image types is weak when using online processing, e.g. JPEG compression, resizing, blur, screenshots and WebP conversion can remove or alter forensic traces. A detector should be tested under these conditions.

For this reason, this thesis intends to fill these gaps by learning a self supervised representation and then by investigating the latent space for detection, generator-family separation and robustness [11], [18].

1.4 Objective

The primary goal of this study is to develop and test a self-supervised representation learning approach to detect the AI-generated images and provide explainability for the detection through latent space analysis.

The specific objectives are:

1. To create a variety of real and synthetic image sets from real photographs and images created by GAN and diffusion models.
2. To train an autoencoder (or contrastive encoder) to learn image embeddings by learning only from indirect real/fake labels.
3. To compare the learned representation with supervised CNN baselines and, if possible, a pre-trained feature extractor like CLIP.
4. To understand the latent space with PCA, UMAP, t-SNE and investigate the clustering of real images, GAN images and diffusion images.
5. To evaluate the quality of the latent space with metrics such as Silhouette Coefficient, cluster compactness, and cross-domain separability as well as standard detection metrics.
6. To test cross-generator generalization by training on selected generators and testing on unseen generators.
7. To test robustness to typical post-processing techniques including JPEG compression, resizing and blur.

1.5 Methodology in Brief

The proposed methodology is an experimental pipeline which is detailed step by step. The design is maintained realistic in the context of a B.Sc. thesis, but provides a clear research contribution. It is based on a workflow instead of a new complex architecture that rely on representation learning, cross-generator testing, and latent-space analysis [11], [17].

During the data stage, real images are chosen from among existing photographic data sets or benchmark sub-sets. Synthetic images are selected from GAN and diffusion-based sources such as ForenSynths, GenImage, CIFAKE, ArtiFact, or similar public datasets [1], [4]. The split is arranged so that at least one generator remains unseen during training.

In the representation-learning stage, an encoder such as ResNet, EfficientNet, ViT, or CLIP image encoder is used. The study may use SimCLR, BYOL, or a SelfCon-style objective depending on implementation resources [10], [19]. The goal is to learn stable features without memorizing only one generator.

After training, the encoder is frozen and each image is converted into a latent embedding. PCA, UMAP, and t-SNE are then used to visualize the embedding space, while Silhouette Coefficient and standard metrics measure separation quality [15], [20].

A lightweight classifier such as logistic regression, linear SVM, or a shallow MLP is trained on top of the frozen embeddings. The result is compared with a fully supervised CNN baseline and, where possible, a CLIP-based feature baseline [8], [9]. The final evaluation tests both seen and unseen generators. It also checks JPEG compression, resizing, WebP conversion, and blur. These tests are included because real online images rarely remain in their original clean form [5], [17].

Proposed Self-Supervised Forensic Representation Learning Workflow

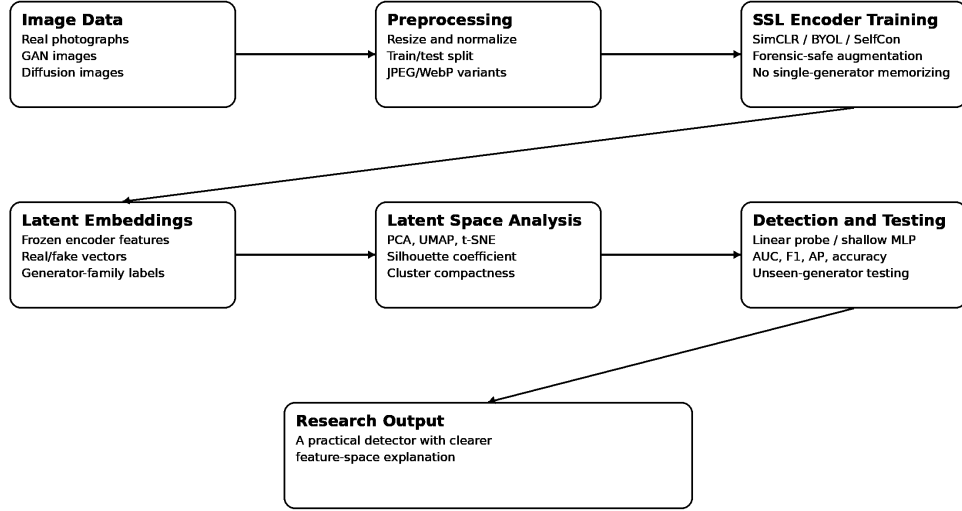


Figure 1.1: Proposed self-supervised forensic representation learning workflow

1.6 Scopes and Challenges

1.6.1 Scopes

- The work focuses on static 2D images. Video deepfakes and audio deepfakes are outside the present scope.
- The main task is image-level classification: real image or Artificial Intelligence-generated image.
- The study considers both GAN-based and diffusion-based image sources where datasets are available.
- The main analysis is done in feature space using embeddings from self-supervised or pretrained models.
- The evaluation includes cross-generator testing and common post-processing tests.

1.6.2 Challenges

- Self-supervised training can require more time and GPU resources than simple supervised training.

- Public datasets may not cover every modern generator, especially proprietary systems.
- If the dataset is not balanced, the model may learn content differences instead of real-versus-generated differences.
- UMAP and t-SNE can show useful visual patterns, but they may distort high-dimensional structure. Quantitative metrics are therefore needed.
- Compression and resizing may remove forensic traces, making robust detection harder.

Chapter 2

Literature Review

2.1 Preliminaries

Artificial Intelligence-generated image detection is the task of deciding whether an image was captured by a real camera or produced by an image generation model. The task is difficult because modern generators try to match the visual distribution of real images [1], [2].

GAN-based generation uses a generator and a discriminator. GAN images may contain artifacts related to up-sampling, texture synthesis, or frequency patterns. Early detectors and co-occurrence methods were built around these traces [6], [21]. Diffusion-based generation creates images through a denoising process. Diffusion images may not contain the same traces as GAN images, so reconstruction-based methods such as DIRE, AEROBLADE, DRCT, FakeInversion, and LaRE2 have become important [3], [22].

Representation learning means learning a useful feature space for images. In this thesis, the feature space is treated as a central part of the detector. A good representation should place related images close together and separate real and generated images when their underlying statistics differ [8], [9].

Self-supervised learning trains a model using signals created from the data itself. SimCLR uses contrastive views, BYOL uses a bootstrap learning process, and SDAIE uses EXIF metadata to learn camera-related traces from real photographs [10], [11].

Latent-space analysis studies the embeddings produced by the model. PCA, UMAP, and t-SNE reduce high-dimensional embeddings for visualization, while Silhouette Coefficient gives a numerical view of cluster separation [15], [16].

2.2 Review of Existing Research

2.2.1 Early supervised and low-level forensic detectors

Wang et al. showed that a ResNet-50 trained on ProGAN images could detect several other CNN-generated images. This result suggested that generated images may share common hidden traces, although the work was mainly centered on CNN-based generators [1].

Nataraj et al. used co-occurrence matrices to capture pixel-level dependencies in GAN images. Later work on up-sampling artifacts also showed that local pixel relationships can carry important forensic information [6], [21].

CIFAKE provided a practical real-versus-generated classification setting and included explainability through visual tools. However, CIFAKE alone is not enough to represent modern high-resolution text-to-image and diffusion outputs [23].

2.2.2 Dataset diversity and benchmark design

GenImage, ArtiFact, WildFake, and ImagiNet show that dataset design is a major part of synthetic image detection. These datasets include different content types, generators, and image sources, which helps reduce overly narrow evaluation [4], [24]. ImagiNet is especially useful for this thesis because it shows that content balance matters. A detector trained without enough variety in faces, paintings, photos, or miscellaneous images may learn category bias instead of forensic traces [17].

Community Forensics takes another direction by using thousands of generators to improve generalization. This supports the idea that a detector should be evaluated beyond a single known generator [13].

2.2.3 Text-to-image and diffusion-specific detection

DE-FAKE studied detection and attribution for text-to-image models using image-only and hybrid image-text features. It also showed that prompt-image relation can be useful for detecting generated images [2].

DIRE, AEROBLADE, DRCT, FakeInversion, and LaRE2 focus more directly on diffusion-generated images. These methods use reconstruction error, inversion behavior, or contrastive reconstruction to expose diffusion-related traces [3], [25].

Synthbuster and spectral learning methods also show that frequency or spectral behavior can remain useful for diffusion image detection. The limitation is that some methods may be strong for one generator family but less complete for all types of generated images [7], [26].

2.2.4 Universal, zero-shot, and multimodal representations

Universal fake detectors and CLIP-based methods show that strong pretrained representations can transfer better to unseen generators. These works are important because they treat the feature space as a key part of generalization [8], [9].

Zero-shot detection, language-guided contrastive learning, MiraGe, image-adaptive prompt learning, and generator-aware prototypes all move the field away from simple closed-set binary classification [14], [27].

These papers also show one limitation: large pretrained or multimodal models can be strong, but their internal feature behavior is not always easy to explain. This supports the need for a separate latent-space analysis in the proposed thesis [28], [29].

2.2.5 Self-supervised and semi-supervised directions

Self-supervised anomaly detection for generated faces shows that a model can learn real-image structure and treat generated images as anomalies. This idea is close to the present thesis because it reduces direct dependence on fake-image labels [18].

SDAIE uses EXIF metadata as a self-supervised signal. Instead of learning fake artifacts, it learns camera-related photographic traces and then detects images that do not fit the real-photo distribution [11].

Semi-supervised detection, generator-aware prototypes, spectral learning, and factor-separation methods show that the field is moving toward more general and less label-dependent detection. These works support a realistic novelty focused on representation quality, not on building a very large new model [30], [31].

2.3 Summary of Key Findings

The reviewed papers show that the field is moving from simple supervised classification toward broader representation-based detection. The most useful findings and limitations are summarized below.

Table 2.1: Key findings and limitations from reviewed literature

Research direction	Works	Main finding	Main limitation	Use in this thesis
Supervised detection	CNN [1], [12], [23]	CNNs can detect many generated images.	They may overfit to known generator artifacts.	Use as baseline, not final explanation.
Pixel/frequency artifacts	[6], [7], [21], [26]	Generated images leave local and spectral traces.	Some cues may disappear under compression or resizing.	Keep forensic safe preprocessing and robustness tests.
Large benchmarks	[4], [5], [13], [17], [24]	Diverse data proves realistic evaluation.	Dataset bias can still remain.	Use balanced real and synthetic subsets where possible.
Diffusion detection	[3], [22], [25], [32], [33]	Reconstruction and inversion errors expose diffusion traces.	Diffusion-specific methods may not cover GANs.	Include both GAN and diffusion sources.
CLIP and multi-modal learning	[2], [8], [9], [14], [27], [28], [29]	Broad pretrained features improve transfer.	The feature behavior is not always explained.	Compare with CLIP and analyze latent space.
Self/semi-supervised methods	[11], [18], [30], [31], [34]	Less label-dependent learning is promising.	Implementation must stay practical and resource-aware.	Use SSL/pretrained features plus linear probe.

2.4 Research Position and Practical Novelty

The proposed thesis is positioned between synthetic image detection and representation analysis. Many papers build a detector, but fewer studies explain how real, GAN, and diffusion images are arranged inside the learned feature space. This thesis keeps that explanation as part of the main objective [11], [17].

The novelty is practical and implementable. The study does not depend on training a very large new foundation model. It uses available backbones, self-supervised or contrastive learning, linear probing, cross-generator testing, and latent-space metrics. This makes the work suitable for the available B.Sc. thesis scope [10], [19]. The expected contribution is a clearer experimental framework: learn embeddings, test them on unseen generators, visualize the feature space, quantify cluster separation, and compare with supervised and pretrained baselines. This gives the work more value than a normal real-versus-fake classifier [8], [9].

Table 2.2: Planned novelty and implementation path

Novelty element	Implementation in this thesis
Self-supervised or contrastive encoder	Use SimCLR, BYOL, SelfCon-style learning, or a pre-trained feature encoder depending on available resources.
Cross-generator evaluation	Train on selected generators and test on unseen GAN or diffusion sources.
Latent-space explanation	Use PCA, UMAP, t-SNE, Silhouette Coefficient, and cluster compactness to explain feature separation.
Robustness testing	Evaluate JPEG compression, resizing, WebP conversion, and blur.
Baseline comparison	Compare with supervised CNN and CLIP/pretrained feature baselines.

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